

Option-Implied Probability Distortions and Stock Return Predictability

Ting-Xuan Wang ^{*} Wei-Yu Kuo [†]

Zong-Wei Yeh [‡] Shih-Kuei Lin [§]

October 30, 2025

Abstract

As retail investors increasingly influence financial markets, understanding how cognitive biases affect asset prices is crucial. This study examines the return predictability and economic value of the curvature parameter of probability weighting function, which captures investors' risk perception through probability distortions. Using an empirical pricing kernel approach with particle filtering, we construct a Probability Weighting Index (*PWI*) and analyze its predictive power. Empirical results suggest that *PWI* exhibits a certain degree of predictive power, provides incremental information beyond traditional predictors, and offers economic value through an investment strategy that outperforms the buy-and-hold approach. Further analysis indicates that *PWI* primarily predicts returns through the discount rate channel, distinguishing it from sentiment-based predictors. Our findings highlight the role of cognitive biases in asset pricing by introducing *PWI* as a potential behavioral-based predictor.

Keywords: probability weighting, return predictability, investment strategy

JEL classification: G11, G12, G23

^{*}Department of Money and Banking, National Chengchi University; 112352015@nccu.edu.tw.

[†]Department of International Business, National Chengchi University; wkuo@nccu.edu.tw.

[‡]Department of Money and Banking, National Chengchi University; 111352502.

[§]Corresponding author, Department of Money and Banking, National Chengchi University; Postal Address: No.64, Sec.2, ZhiNan Rd., Wenshan District, Taipei City 11605, Taiwan; square@nccu.edu.tw.

1 Introduction

In traditional stock return predictability research, scholars have predominantly focused on macro predictors.¹ While these variables capture broad economic conditions, they often overlook the role of individual investor behavior and psychological biases in shaping market dynamics.

Over the past decade, the composition of market participants has shifted significantly, with retail investors playing an increasingly dominant role in equity trading.² According to Bloomberg Intelligence, the share of retail investors in total trading volume has increased from just above 10% in 2011 to more than 22% in 2021 (Salvucci, 2023). While both retail and institutional investors are subject to behavioral finance biases, prior research suggests that retail investors exhibit these biases more strongly.³ Given that retail investors now play an increasingly dominant role in equity trading, their behavioral tendencies may exert a greater influence on stock returns, making investor psychology an increasingly important factor in return predictability. This shift raises an important question: How do cognitive biases, particularly in risk perception, affect stock market dynamics?

Behavioral finance provides a framework for answering this question, emphasizing how investors systematically deviate from rational decision-making due to cognitive biases. One key concept in this literature is the probability weighting function, which describes how individuals distort probabilities when evaluating risk, often overweighting small probabilities and underweighting large ones (Tversky and Kahneman, 1992). The degree of this distortion—captured by the curvature of the probability weighting function—varies across investors and market conditions⁴, making it a potential driver of return

¹There are many macro predictors such as dividend yield (Campbell and Shiller, 1988), interest rates (Fama and French, 1989) and economic uncertainty (Ludvigson and Ng, 2007). More recently, studies have also explored energy-related variables, such as oil prices and oil futures prices, as return predictors Chiang and Hughen (2017); Wang et al. (2019), further reinforcing the dominance of macroeconomic factors in this field.

²Retail investor participation in U.S. equity markets has remained elevated even after peaking in Q1 2021, with retail on Cboe’s EDGX Equities Exchange accounting for 1.3% of total market volume in Q3 2024, reflecting a 176% increase from Q1 2018 and an 85% increase from Q1 2020. The retail Priority Order mechanism on EDGX, which prioritizes retail orders over non-retail orders at the same price level, has enhanced execution quality and reduced time-to-fill for retail investors, further supporting their sustained presence in the market (Cboe, 2024).

³Studies have documented that retail traders are more prone to excessive trading Barber and Odean (2000), the disposition effect (Shefrin and Statman, 1985), and home bias Kumar (2009). In contrast, institutional investors are often constrained by professional risk management practices, investment mandates, and performance benchmarks, which may limit the extent of their psychological distortions (Shleifer and Vishny, 1997).

⁴Charupat et al. (2013) suggests that individual differences such as emotional balance, influence probability weighting. Individuals with higher emotional stability exhibit less distorted probability perceptions.

predictability. This study seeks to examine whether the curvature parameter of the probability weighting function serves as a reliable predictor of stock returns. Additionally, we investigate whether an investment strategy based on this measure outperforms the traditional buy-and-hold strategy.

While prior research has incorporated investor sentiment as an irrational predictor (Brown and Cliff, 2005; Huang et al., 2015), sentiment-based proxies—such as survey responses or media tone—may fail to fully capture the cognitive distortions underlying investment decisions. Unlike sentiment measures, probability weighting provides a more direct representation of how investors perceive risk. By addressing the two key research questions—(1) Does the curvature parameter of the probability weighting function serve as a reliable predictor of stock returns? (2) Can an investment strategy based on curvature outperform the traditional buy-and-hold strategy?—this research not only contributes to the growing literature on behavioral finance and return predictability but also evaluates the practical implications of curvature-based trading strategies.

To examine the predictive power of the curvature parameter in the probability weighting function, we first need to estimate the function itself. In the literature, probability weighting functions are estimated using either an experimental approach or a market-based approach. The experimental approach infers the probability weighting function by analyzing individuals' choices in controlled decision tasks with known probabilities and payoffs Tversky and Kahneman (1992); Gonzalez and Wu (1999)⁵. In contrast, the market-based approach estimates the probability weighting function using market data, primarily from options. Polkovnichenko and Zhao (2013) employ a nonparametric method that combines risk-neutral and physical distributions.⁶ By comparing these densities, they infer the implied probability weighting function. Similarly, Shi et al. (2023) estimate the curvature parameter using the empirical pricing kernel approach (Rosenberg and Engle, 2002), specifying the probability weighting function in the form proposed by Prelec (1998).

Building on Shi et al. (2023), our approach integrates the empirical pricing kernel method with the particle filter framework to dynamically estimate the curvature parameter. This integration allows for more accurate filtering and smoothing distributions compared to

Polkovnichenko and Zhao (2013) find that the shape of the probability weighting function is time-varying, sometimes transitioning to an S-shape, suggesting that investors' probability distortions fluctuate with market conditions.

⁵Tversky and Kahneman (1992) estimate the curvature parameter (α) to be 0.61 using a single-parameter function, indicating an inverse S-shape where small probabilities are overweighted and large probabilities are underweighted. Gonzalez and Wu (1999) employ a more flexible two-parameter model and estimate the curvature parameter (γ) to be 0.44.

⁶Polkovnichenko and Zhao (2013) extracts the risk-neutral density from option prices using the Breeden and Litzenberger (1978) methodology and estimating the physical density from historical S&P 500 returns via an EGARCH model.

traditional estimation techniques [Lopes and Tsay \(2011\)](#). To implement the pricing kernel approach, we first derive a pricing kernel under the Rank-Dependent Expected Utility (RDEU) preferences. While [Baele et al. \(2019\)](#) derive a pricing kernel under cumulative prospect theory, our model focuses solely on the probability weighting component rather than both the value function and probability weighting. The pricing kernel we obtain closely aligns with those derived in [Shi et al. \(2023\)](#) and [Polkovnichenko and Zhao \(2013\)](#)⁷. Finally, we extract the probability weighting function from option data using the derived pricing kernel and take the reciprocal of the curvature parameter, define as probability weighting index (*PWI*).

Next, we evaluate the return predictability of our proposed predictor, following established methodologies in the return predictability literature ([Azimi et al., 2024](#); [Wang et al., 2019](#)). Our analysis consist of three key components: in-sample tests, out-of-sample tests, and an assessment of economic significance.

For the in-sample analysis, we employ univariate regressions to directly assess the predictive power of our proposed predictor over the full sample period. Additionally, we conduct bivariate regressions to examine whether our predictor provides incremental predictive information beyond existing predictors documented in prior literature. The results suggest that our predictor exhibits significant predictive power and outperforms most of the 14 predictors identified in [Goyal and Welch \(2008\)](#) (hereafter, GW predictors)⁸. Furthermore, it provides additional predictive information beyond most of the GW predictors, highlighting its unique contribution to return predictability. For the out-of-sample analysis, we employ both the split-sample and expanding-window approaches to evaluate the robustness of our predictor’s forecasting performance relative to alternative predictors. These tests help mitigate concerns of in-sample overfitting and provide a more realistic assessment of predictive ability in real-time forecasting environments. While none of the candidate predictors, including our own, consistently outperform the historical mean benchmark, our predictor demonstrates superior predictive ability relative to the other predictors, as reflected in its higher out-of-sample R^2 (OOS R^2).

Beyond statistical significance, we assess the economic significance of our predictor by constructing an investment strategy based on the optimal portfolio allocation framework of [Merton \(1969\)](#)⁹. Our findings suggest that even if the statistical tests do not provide

⁷In [Polkovnichenko and Zhao \(2013\)](#) and [Shi et al. \(2023\)](#), the pricing kernel under RDEU is the product of the marginal utility and the derivative of probability weighting.

⁸[Goyal and Welch \(2008\)](#) conduct a comprehensive analysis of return predictability using 14 commonly studied macroeconomic and financial predictors. Their findings challenge the robustness of predictive regressions, showing that many predictors fail to outperform a simple historical mean model in out-of-sample tests.

⁹[Merton \(1969\)](#) introduces a continuous-time portfolio selection model where investors dynamically allo-

overwhelming support for its predictive power, our predictor can still enhance investment decisions by improving portfolio performance. To quantify its economic value, we evaluate portfolio performance using certainty equivalent return (CEQ) ¹⁰, following [Chiang and Huguen \(2017\)](#); [Wang et al. \(2019\)](#). The results indicate that an investment strategy based on our predictor achieves higher CEQ than most GW predictors across different levels of risk aversion.

As an extension, we employ the forecast combination approach of [Rapach et al. \(2010\)](#) to further assess whether our proposed predictor enhances return forecasts. Specifically, we compare the predictive performance of a model that combines the GW predictors with one that incorporates both the GW predictors and our proposed predictor. To ensure robustness, we implement five different forecast combination methods. The results consistently show that including our predictor improves CEQ across all combination approaches, further reinforcing its relevance in return forecasting.

Classical asset pricing models suggest that stock prices are driven by both expected future cash flows and the discount rates applied to them. Based on this framework, we explore the predictive channel of our predictor by investigating whether it forecasts expected cash flows, discount rates, or both, following [Huang et al. \(2015\)](#). Specifically, we use the dividend-price ratio (DP) and long-term yield (LTY) as proxies for the discount rate, and dividend growth (DG) and earnings growth (EG) as proxies for expected cash flows ¹¹. Predictive regression estimates suggest that our predictor primarily forecasts returns through the discount rate channel, rather than through expected cash flow.

Our findings contribute to the growing literature on probability weighting in finance and return predictability in three key ways. First, we introduce a novel approach to estimating the probability weighting function by combining the particle filter method with the empirical pricing kernel. This Bayesian estimation framework allows for a dynamic assessment of probability weighting while accounting for its time-varying nature and autocorrelation. Second, we propose the Probability Weighting Index (*PWI*) as a predictor of stock returns, capturing the extent to which investors overweight or underweight small probabilities. Our results show that *PWI* exhibits stronger predictive power than traditional macroeconomic predictors documented in prior literature. Third, *PWI* demonstrates eco-

cate between a risk-free and a risky asset to maximize lifetime utility. The framework provides a foundation for intertemporal asset allocation under uncertainty and has been widely used in return predictability studies.

¹⁰Certainty Equivalent Return (CEQ) measures the guaranteed return an investor would require to be indifferent between holding a risky portfolio and receiving a risk-free return. It is commonly used to evaluate the economic value in asset allocation.

¹¹[Huang et al. \(2015\)](#) employs the dividend-price ratio (DP) as a proxy for the discount rate, while using dividend growth (DG), earnings growth (EG), and GDP growth as proxies for expected cash flows.

conomic significance by generating higher CEQ than the buy-and-hold strategy, underscoring its practical value in investment decision-making. Our study is most closely related to [Azimi et al. \(2024\)](#), which examines the aggregate market predictability of the α parameter under the NEO-EU model framework. Their approach focuses on how investors distort the probabilities of extreme events, interpreting α as a measure of optimism, whereas our framework employs the traditional probability weighting function (PWF), capturing risk perception rather than optimism alone. ¹²

The remainder of the paper is structured as follows. Section 2 reviews the literature on probability weighting in finance and time-series return predictability. Section 3 outlines our methodological framework. Section 4 presents the empirical findings, evaluating the predictive power of *PWI* and exploring its underlying economic mechanisms. Section 5 concludes the study.

2 Literature Review

2.1 Probability weighting function

The probability weighting function (PWF) has been widely studied in decision theory and behavioral finance, as it captures how individuals distort objective probabilities when making decisions under uncertainty. One of the foundational insights in this literature is that individuals tend to overweight small probabilities and underweight large probabilities, leading to an inverse S-shaped probability weighting curve. This behavioral pattern, first introduced in Prospect Theory ([Tversky and Kahneman, 1992](#)), has significant implications for risk perception and financial decision-making. However, the conceptual foundation of probability weighting can be traced further back to Rank-Dependent Expected Utility (RDEU) Theory, first introduced by [Quiggin \(1982\)](#) as an extension of Expected Utility Theory (EUT). Unlike EUT, which assumes that individuals evaluate uncertain prospects using objective probabilities, RDEU allows for nonlinear probability transformations, where decision-makers weight probabilities asymmetrically based on rank-dependent transformations. This modification enables RDEU to address well-documented violations of EUT, such as the Allais Paradox, by accounting for observed

¹²The NEO-EU model selectively distorts the probabilities of extreme events, where the α parameter determines whether positive (good) or negative (bad) extreme outcomes are overweighted. A positive α reflects optimism, leading to overweighting of positive extreme events and underweighting of negative ones, whereas a negative α reflects pessimism, inducing the opposite probability distortion. In contrast, the traditional probability weighting function (PWF) applies a continuous curvature that captures risk perception across the full probability distribution, rather than focusing solely on extreme events.

risk attitudes across different probability levels. RDEU maintains a single global utility function but modifies probability weighting in a way that ensures risk preferences depend on the ranking of outcomes rather than absolute probabilities, preserving stochastic dominance while incorporating behavioral distortions.

The functional form of the PWF has been extensively analyzed, with various specifications proposed to model probability distortions. One widely used representation is the [Prelec \(1998\)](#) compound invariant function, which takes the form

$$w(p) = \exp(-(\ln p)^\alpha), \quad (1)$$

where the parameter α determines the curvature of probability distortions. This formulation ensures that the probability transformation remains consistent across different probability levels. Another widely adopted model is the [Tversky and Kahneman \(1992\)](#) function, given by

$$w(p) = \frac{p^\gamma}{(p^\gamma + (1-p)^\gamma)^{1/\gamma}}, \quad (2)$$

which allows for flexible probability distortions depending on the value of γ . While these probability weighting functions capture the general inverse S-shaped distortion pattern, later contributions have refined these models to accommodate additional features. For example, the neo-additive probability weighting function ([Chateauneuf et al., 2007](#)) introduces a structure that maintains probability distortions primarily at the extremes of the distribution, while leaving intermediate probabilities largely unchanged. These variations provide different ways of capturing probability distortions, with empirical studies often calibrating their functional forms to best fit observed behavior.

The estimation of probability weighting functions has been approached using both experimental and market-based methods. Traditional experimental approaches have relied on controlled decision tasks in which subjects choose between lotteries with known probabilities, allowing researchers to infer the shape of the probability weighting function by observing their choices. These methods have consistently validated the inverse S-shape of the function, confirming that individuals tend to overweight rare events while underweighting more probable outcomes ([Tversky and Kahneman, 1992](#); [Gonzalez and Wu, 1999](#)). However, concerns about external validity have led researchers to explore market-based estimation techniques, particularly using financial market data. One prominent approach involves estimating PWFs using option data, as option prices inherently reflect investor's probability perceptions. Studies such as [Polkovnichenko and Zhao \(2013\)](#) have derived nonparametric estimates of PWFs using S&P 500 index options, finding strong

evidence of probability distortions in market pricing. Similarly, [Chen et al. \(2024\)](#) proposed an alternative method that infers PWFs through option pricing bounds, showing that probability distortions persist across different market conditions. A recent and notable contribution by [Shi et al. \(2023\)](#) employs a structural asset pricing approach to estimate probability weighting through empirical pricing kernels. Their study leverages Generalized Method of Moments (GMM) estimation on option data to obtain time-varying estimates of the probability weighting curvature parameter (α), which directly captures how investors distort probabilities in financial markets.

The probability weighting function has been widely applied in finance to explain various asset pricing anomalies, return predictability, and portfolio allocation decisions. One key area of application is asset pricing models, where probability weighting has been used to refine traditional frameworks. [Shi et al. \(2023\)](#) incorporated probability weighting into a three-moment CAPM, demonstrating that probability distortions influence both beta and coskewness pricing. Similarly, [Driessen et al. \(2025\)](#) introduced the Π -CAPM, which extends the traditional capital asset pricing model by integrating probability weighting into investor's risk evaluations. Their findings suggest that probability distortions play a significant role in shaping asset prices and risk premia, affecting the pricing of assets with skewed return distributions.

Beyond asset pricing, probability weighting has been linked to option mispricing and variance risk premiums. Research has shown that traditional models often struggle to explain the pricing of out-of-the-money (OTM) options, as their implied probabilities appear systematically distorted. By incorporating probability weighting into option pricing models, researchers have provided explanations for the non-monotonicity of the pricing kernel and the overpricing of downside protection (puts) relative to upside potential (calls) ([Baele et al., 2019](#)). These findings align with market observations that investors often demand excessive risk premiums for low-probability, high-impact events.

In the broader equity market, probability weighting has also been used to explain return predictability and market anomalies. Some studies suggest that the curvature of the probability weighting function may serve as a predictor for future stock returns, capturing shifts in investor sentiment over time ([Azimi et al., 2024](#)). Additionally, probability weighting has been proposed as an explanation for the beta anomaly, where high-beta stocks underperform relative to traditional CAPM predictions. [Shi et al. \(2023\)](#) found that probability distortions lead to a flat or downward-sloping-beta-return relationship, challenging the classical risk-return tradeoff.

Investor behavior biases such as the disposition effect and lottery-like stock preferences have also been linked to probability weighting. [Meng and Weng \(2018\)](#) argued that

CPT-based models incorporating probability weighting provide a more comprehensive explanation of the disposition effect, where investors tend to hold losing stocks too long and sell winning stocks too early. Similarly, [Barberis and Huang \(2008\)](#) demonstrated that probability weighting contributes to lottery-like stock demand, where investors overpay for stocks with extreme upside potential despite their negative expected returns. These findings highlight how low probability distortions can drive systematic mispricing in financial markets.

Finally, in portfolio allocation and market participation, probability weighting has been found to influence investor decisions in dynamic settings. [De Giorgi and Legg \(2012\)](#) showed that probability weighting, particularly when combined with narrow framing, can lead to stock market non-participation, as investors perceive equities as excessively risky due to distorted probability perceptions. [He and Sun \(2023\)](#) extend this analysis by incorporating time-inconsistent probability weighting into dynamic portfolio selection, demonstrating that agents with probability distortions adjust their investment strategies in response to perceived changes in risk. These studies suggest that probability weighting has important implications not only for asset pricing but also for how investors allocate capital over time.

2.2 Return predictability

The predictability of aggregate market returns has long been a central issue in financial economics, with research exploring various macroeconomic and financial predictors. Early studies, such as [Fama and Schwert \(1977\)](#) and [Fama and French \(1989\)](#), documented that valuation ratios, including the dividend-price ratio, earnings-price ratio, and book-to-market ratio, exhibit significant predictive power for future returns. These findings challenged the efficient market hypothesis and suggested that expected stock returns are time-varying. Similarly, [Campbell and Shiller \(1988\)](#) demonstrated that high dividend yields tend to forecast higher future stock returns, providing further empirical support for return predictability. Moreover, term spreads and short-term interest rates were found to be significant predictors of future equity risk premia, as documented by [Campbell \(1987\)](#).

Despite these early findings, the robustness of return predictability has been widely debated, particularly regarding out-of-sample forecasting performance. [Goyal and Welch \(2008\)](#) conducted a comprehensive study evaluating the predictive power of various financial and macroeconomic variables, concluding that many commonly used predictors fail to outperform a simple historical mean forecast. Their results raised concerns about data mining, structural breaks, and model instability, casting doubt on the economic sig-

nificance of return predictors. Similarly, [Stambaugh \(1999\)](#) highlighted the statistical challenges associated with predictive regressions, noting that small-sample biases often exaggerate the apparent predictive ability of financial variables.

In response to these criticisms, several studies have proposed methods to improve return predictability. [Campbell and Thompson \(2008\)](#) suggested imposing theoretical constraints—such as requiring predictor coefficients to have the correct economic sign—to enhance the out-of-sample performance of predictive regressions. Their findings indicate that valuation ratios, when appropriately constrained, retain significant predictive power. [Cochrane \(2008\)](#) defended return predictability by demonstrating that if returns were truly unpredictable, then dividend growth must be predictable—a claim that is not supported by empirical evidence. Furthermore, [Lewellen \(2004\)](#) examined financial ratios as return predictors and found that while some variables lose significance out-of-sample, others remain robust under different estimation techniques.

Beyond traditional macroeconomic predictors, researchers have explored novel approaches to stock return forecasting. [Rapach et al. \(2010\)](#) proposed combining multiple predictive models, arguing that aggregating information across different predictors improves forecasting robustness. Their findings suggest that a combination of valuation ratios, interest rate spreads, and macroeconomic indicators provides superior predictive performance compared to individual regressions. [Boudoukh et al. \(2007\)](#) introduced payout yield as an alternative predictor, which incorporates both dividends and share repurchases, demonstrating that it has stronger predictive power than the traditional dividend yield.

Another key area of research is the post-publication decline in predictability, as examined by [McLean and Pontiff \(2016\)](#). Their study analyzed 97 previously documented return predictors and found that the predictive power of many variables declines significantly after publication, suggesting that market participants adapt to newly discovered anomalies. This raises important questions about whether return predictability reflects genuine risk premia variation or temporary market inefficiencies that dissipate as investors arbitrage them away.

Overall, the literature on return predictability highlights both the challenges and opportunities associated with forecasting stock returns. While traditional predictors such as valuation ratios and interest rates remain widely studied, their forecasting power is often sensitive to sample selection, structural breaks, and economic techniques. Recent advances in forecast combination models, theoretical constraints on regressions, and alternative predictors such as payout yield offer promising avenues for improving the robustness of return forecasting. However, the debate on whether return predictability primarily re-

flects rational risk premia or behavioral mispricing remains an open question, motivating further research in this field.

3 Methodology

3.1 Asset Pricing Model

3.1.1 Rank Dependent Expected Utility

Rank Dependent Expected Utility (RDEU) is an extension of the traditional expected utility framework designed to account for observed deviations from rational decision-making under risk. In standard expected utility theory, the utility of an outcome is weighted directly by its objective probability. However, empirical evidence shows that individuals often distort probabilities—overweighting low-probability events and underweighting high-probability events. RDEU addresses this by introducing a probability weighting function that transforms the cumulative probabilities of ranked outcomes.

More specifically, in RDEU, outcomes are first ordered from worst to best, and then the decision-maker's evaluation of each outcome is determined not only by its utility but also by a weight derived from the transformed cumulative probability. Mathematically, suppose there are n outcomes denoted by $x_1, x_2, \dots, x_n$ (arranged in increasing order) with probabilities $p(x_1), p(x_2), \dots, p(x_n)$ and $\tilde{F}(x_i)$ represents the decumulative probability up to outcome x_i :

$$\tilde{F}(x_i) = 1 - w(1 - F(x_i)), \quad (3)$$

where $w(\cdot)$ is the probability weighting function that transfers objective probabilities into decision weights with the restrictions $w(0) = 0$ and $w(1) = 1$, $F(\cdot)$ is the actual cumulative distribution function. The RDEU value is calculated as

$$RDEU = \sum_i [w(\tilde{F}(x_i)) - w(\tilde{F}(x_{i+1}))] u(x_i), \quad (4)$$

where $u(x_i)$ is the utility for outcome i . In Equation 3.2, $w(\tilde{F}(x_i)) - w(\tilde{F}(x_{i+1}))$ is the decision weight for outcome x_i with probability p_i , we denote the decision weight as $\pi(p_i)$ and Equation 3.2 can be written as

$$RDEU = \sum_i \pi(p_i) u(x_i), \quad (5)$$

This formulation preserves first-order stochastic dominance and is particularly effective at

explaining anomalies such as the Allais paradox, where individual's choices deviate from those predicted by standard expected utility theory.

3.1.2 Pricing Kernel

To analyze the impact of subjective probability distortions on the evaluation of risky prospects and asset pricing, we adopt the RDEU framework to model investor behavior. Specifically, we consider an investor with RDEU preferences, who evaluates risky prospects based on a probability weighting function. The decision-making process follows Equation 3.3, which accounts for distortions in perceived probabilities.

We consider an one-period model starting from time 0 and end at time T in a simple market with an equity index that pays x_i at time T in state i , which occurs with probability p_i . The investor problem is to determine the share of wealth to be invested in the risky asset, ψ^E , such that the RDEU value is maximized. Given the investor's initial wealth W_0 , her terminal wealth in state i is given by

$$W_{T,i} = [R_f + \psi^E (R_i - R_f)] \cdot W_0, \quad (6)$$

where $R_i = x_i/s_0$ with s_0 the time 0 equity price, R_f is the risk-free rate and ψ^E is the share of wealth invested in equity index at time 0. We write the utility under investor's terminal wealth, plug Equation 3.4 into Equation 3.3 and obtain

$$RDEU = \sum_i \pi(p_i) \cdot u \left([R_f + \psi^E (R_i - R_f)] \cdot W_0 \right) \quad (7)$$

Equation 3.5 is the RDEU value that investor tries to maximize by deciding the optimal share of wealth ψ^E to invest in equity index.

In our model, we specify the utility as Constant Relative Risk Aversion (CRRA) utility function. As for the probability weighting function in $\pi(p_i)$, we use the Prelec's weighting function, which is commonly used in the literature. The form of CRRA utility function and Prelec's weighting function are as follow:

$$u(x) = \frac{x^{1-\gamma}}{1-\gamma}, \quad w(x) = \exp \left(-(\log p)^{\frac{1}{\alpha}} \right),$$

where x is the outcome, in our case, investor's terminal wealth. γ is the risk aversion coefficient, p is the probability for outcome x and α is the curvature parameter of the probability weighting function, which controls the degree of distortion in subjective probability perception. Specifically, α determines how individuals overweight or underweight prob-

abilities relative to their objective values. The Prelec's probability weighting function is plotted in Figure 3.1, with different levels of α .

[Insert Figure 1 here]

To maximize the RDEU value, we take the derivative of Equation 3.5 with respect to ψ^E , which yields the first-order condition (FOC)

$$\sum_i [R_f + \psi^E(R_i - R_f)]^{-\gamma} (R_i - R_f) \pi(p_i) = 0, \quad (8)$$

where initial wealth W_0 is normalized to 1. In equilibrium, the market must clear, and the investor must hold the entire supply of equity, which implies $\psi^E = 1$. The FOC becomes

$$\begin{aligned} 0 &= \sum_i R_i^{-\gamma} \pi(p_i) (R_i - R_f) \\ &= \sum_i p_i R_i^{-\gamma} \frac{\pi(p_i)}{p_i} (R_i - R_f) \\ &= E \left[(R - R_r) R^{-\gamma} \frac{\pi(p)}{p} \right] \\ &\equiv E \left[(R - R_r) \tilde{m} \right] \end{aligned} \quad (9)$$

where $\tilde{m} = R^{-\gamma} \frac{\pi(p)}{p}$ represents the pricing kernel under RDEU preferences. This pricing kernel can be used to price various assets, including options. To examine its behavior under different market conditions, we plot the pricing kernel as a function of the market return. Figure 3.2 illustrates the relationship between the pricing kernel and the market return under different market conditions.

[Insert Figure 2 here]

When $\alpha < 1$, an investor with RDEU preferences tends to overweight tail events, leading to a greater willingness to pay higher prices in extreme states. This behavior results in a U-shaped pricing kernel. Conversely, when $\alpha > 1$, tail events are underweighted, producing the opposite effect—a bell-shaped pricing kernel. This contrast with the pricing kernel derived under Constant Relative Risk Aversion (CRRA) utility.

3.2 Estimating Curvature

3.2.1 Empirical Pricing Kernel

To estimate the curvature parameter of the probability weighting function under RDEU preferences, we employ an empirical pricing kernel (EPK) approach proposed by [Rosenberg and Engle \(2002\)](#). The EPK is derived from observed option prices and an estimated return density, enabling the identification of a time-varying pricing kernel that reflects investor risk attitudes across different market conditions.

The pricing kernel, denoted as $\tilde{m}(\theta, X)$, characterizes investor's preferences for state contingent payoffs and is defined by the standard risk-neutral pricing relationship:

$$P_t = E_t [\tilde{m}(\theta_{t+1}, X_{t+1}) X_{t+1}], \quad (10)$$

where P_t is the current price of the asset, X_{t+1} represents the asset payoff at time $t + 1$, and $\tilde{m}(\theta, X)$ is the pricing kernel parameterized by θ , which governs investor preferences under different states characterized by X . Assuming a finite number of future states with known probabilities, the theoretical asset price can be computed numerically using Equation 3.8. The EPK at time t is then estimated by minimizing the squared deviation between observed and theoretical prices:

$$\min_{\theta_t} \sum_i [P_{i,t} - \hat{P}_{i,t}(\theta_t)]^2, \quad (11)$$

where $P_{i,t}$ is the observed price of the i th asset at time t , and $\hat{P}_{i,t}(\theta_t)$ is the corresponding theoretical price derived from the pricing model. In this study, we extract the EPK from observed option prices. Specifically, the theoretical price of a European call option is given by:

$$C(\theta) = E[\tilde{m}(\theta, X)(S_T - K_C)^+], \quad (12)$$

where θ represents the preference parameter in the pricing kernel, X denotes the return under different states, S_T is the underlying asset price at maturity T , and K_C is the strike price of the call option. Similarly, the theoretical price of a European put option is given by:

$$P(\theta) = E[\tilde{m}(\theta, X)(K_P - S_T)^+], \quad (13)$$

where K_P is the strike price of the put option.

To mitigate scale-related issues in option prices, some studies minimize the difference between observed and theoretical implied volatilities rather than the difference between

observed and theoretical option prices. Once theoretical option prices are obtained, the corresponding implied volatilities can be derived using the Black-Scholes pricing formula. The implied volatilities for call option and put options are denoted as $IV_C(\theta, K_C)$ and $IV_P(\theta, K_P)$ respectively. The estimation problem is then reformulated as:

$$\min_{\theta} \sum_i [IV(\theta, K) - IV^{real}(\theta, K)]^2, \quad (14)$$

where $IV(\theta, K)$ represents the theoretical implied volatility for either call or put options, and $IV^{real}(\theta, K)$ denotes the corresponding observed implied volatility from market data.

Rather than estimating the EPK at each time t directly by solving the optimization problem in Equation 3.12, we adopt a particle filter approach. In this framework, the discrepancy between theoretical and observed implied volatilities for each option is incorporated as the observation equation in a state-space model. The details of the particle filtering methodology will be elaborated in Section 3.2.3.

3.2.2 Physical Distribution

In the previous section, we discussed the computation of the EPK under the assumption of a finite number of future states with known probabilities. More specifically, different market states are characterized by distinct market returns, denoted as R . It is widely recognized in the literature that stock returns do not follow a normal distribution. Consequently, models such as GARCH and asymmetric GARCH are frequently used to capture the fat tails and skewness observed in stock return distributions.

In this study, we fit the daily stock return data using various volatility models, including ARCH, GARCH, and EGARCH. To determine the most appropriate model, we select the one that maximizes the log-likelihood, as this criterion identifies the model that best fits the observed stock return data. To formally define the volatility models used in this study, we present the mathematical specifications of the ARCH, GARCH, and EGARCH models as follows. Each of these models describes the conditional variance σ_t^2 of stock returns, incorporating different levels of complexity in capturing time-varying volatility dynamics.

ARCH(1) Model

The ARCH(1) model assumes that the conditional variance depends on past squared

innovations. Its specification is given by:

$$\begin{aligned} \log(S_t) &= \log(S_{t-1}) + \mu + \sqrt{h_t}z_t, \quad z_t \sim N(0, 1), \\ h_t &= \omega + \alpha h_{t-1}z_{t-1}^2, \quad \omega > 0, \quad \alpha \geq 0, \end{aligned} \quad (15)$$

GARCH(1,1) Model

The GARCH(1,1) model extends the ARCH model by incorporating lagged conditional variance, allowing for greater persistence in volatility:

$$\begin{aligned} \log(S_t) &= \log(S_{t-1}) + \mu + \sqrt{h_t}z_t, \quad z_t \sim N(0, 1), \\ h_t &= \omega + \alpha h_{t-1}z_{t-1}^2 + \beta h_{t-1}, \quad \omega > 0, \quad \alpha \geq 0, \end{aligned} \quad (16)$$

EGARCH(1,1)

Unlike ARCH and GARCH models, which model variance directly, the EGARCH model specifies the logarithm of variance, ensuring that h_t remains positive without requiring non-negativity constraints on parameters. The EGARCH(1,1) model is given by:

$$\begin{aligned} \log(S_t) &= \log(S_{t-1}) + \mu + \sqrt{h_t}z_t, \quad z_t \sim N(0, 1), \\ \log(h_t) &= \nu + \beta \log(h_{t-1}) + \zeta(|z_{t-1}| - E[|z_{t-1}|]) + \kappa z_{t-1}, \end{aligned} \quad (17)$$

Here, ζ captures the asymmetric impact of past absolute standardized innovations, while κ represents the leverage effect, allowing negative shocks to have a different impact on volatility than positive shocks.

After identifying the model that best fits stock returns, we use it to simulate stock returns at the option's expiration at time t . Specifically, at the end of each month, we collect the past 1250 daily S&P 500 index return observations and estimate the model parameters via maximum likelihood estimation (MLE). Using these estimated parameters, we generate simulated future stock returns. To obtain a smooth representation of the simulated return distribution, we apply kernel density estimation (KDE) with bandwidth equal to $0.9N^{-1/5}\sigma$, where σ is the return standard deviation and N is the number of replications. Next, we define the return range as $[-0.5, 0.8]$ and discretize it into 200 evenly spaced grid points. For each grid point, we compute the cumulative distribution function (CDF) based on the simulated return distribution. This discretized CDF representation facilitates the subsequent calculation of decision weights under the probability weighting function framework.

3.2.3 Particle Filter

In this study, we use particle filter approach rather than the commonly used GMM approach to estimate the EPK as well as the curvature parameter of probability weighting function. The particle filter estimates latent states by propagating weighted particles over time, updating them with observations, and resampling to prevent degeneracy in non-linear models. The key advantages of the particle filter is its ability to account for the autocorrelation of latent variables, which is crucial for modeling dynamic systems. To formally define the particle filter approach used in this study, we now outline the key steps involved in the filtering procedure.

1. Set up the state-space model. The state-space model consists of two key components: the state equation and the observation equation, given by

$$\begin{aligned} \text{state equation} : x_t &= \mu + \phi x_{t-1} + v, \quad v \sim (0, Q) \\ \text{observation equation} : IV_{i,t}^{real} &= IV_{i,t}^{theoretical} + \omega, \quad \omega \sim (0, R) \end{aligned} \quad (18)$$

In Equation 3.16, the state equation characterizes the evolution of the latent state variable over time, while the observation equation describes the relationship between observed data and the hidden state. In our framework, the latent states x_t correspond to the curvature parameter of the probability weighting function. The parameter ϕ represents the autocorrelation coefficient, governing the persistence of the state process, while v and ω denote the process noise and observation noise, respectively. It is important to note that the IV_t^{real} and $IV_t^{theoretical}$ are vectors in this study, as we utilize multiple options with different strike prices, leading to multiple observations at each time step.

2. For $i \in \{1, 2, \dots, N\}$, the time t curvature parameter α_t is simulated from the state equation:

$$\alpha_t = \mu + \phi \alpha_{t-1} + v, \quad v \sim (0, Q) \quad (19)$$

3. For each particle, we calculate the theoretical price and implied volatility of each option using Equation 3.8 and the simulated distribution of stock returns.
4. Update the importance weight for each particle by the theoretical implied volatility calculated and the observation equation. To be more specific, suppose the theoretical implied volatility for the N th particle is IV_N , while the observed implied volatility is IV_N^{real} . To incorporate observational uncertainty, we evaluate the probability density

of IV_N under a multivariate normal distribution centered at IV_N^{real} with a covariance matrix determined by the noise process. Mathematically, this is expressed as:

$$w_{N,t} \propto w_{N,t-1} \cdot p(IV_N | IV_N^{real}, R), \quad (20)$$

where $p(IV_N | IV_N^{real}, R)$ represents the probability density function of a normal distribution with mean IV_N^{real} and variance R , which characterizes the observational noise. This likelihood function quantifies the compatibility between the simulated implied volatility and the observed market data.

The updated importance weight $w_{N,t}$ is then obtained by multiplying the previous time step's weight $w_{N,t-1}$ by this likelihood value. This ensures that particles whose theoretical implied volatilities align more closely with the observed values receive higher weights, while those with greater discrepancies are downweighted.

5. For $i \in \{1, 2, \dots, N\}$, compute the normalized weights $\tilde{w}$:

$$\tilde{w}_{i,t} = \frac{w_{i,t}}{\sum_{k=1}^N w_{k,t}}. \quad (21)$$

6. Resample the particles. In the particle filtering framework, resampling is a crucial step that prevents particle degeneracy by eliminating particles with low weights and replicating those with higher weights. We have already normalized the updated weights in the previous step, the effective sample size (N_{eff}) is then computed as

$$N_{eff} = \frac{1}{\sum \tilde{w}_i^2}, \quad \forall i \in \{1, 2, \dots, N\}, \quad (22)$$

where $\tilde{w}_i$ represents the standardized weight of the i th particle. A smaller N_{eff} indicates that a few particles dominate the distribution, leading to weight collapse. To mitigate this issue, we apply resampling whenever N_{eff} falls below one-third of the total number of particles.

Resampling is performed using systematic resampling, where particles are sampled with replacement in proportion to their weights. This ensures that highly probable particles are more frequently retained, while less likely ones are discarded. After resampling, all particles are assigned equal weights to maintain numerical stability, allowing the particle filter to continue tracking the latent state effectively.

7. The final step of the particle filter is to compute the estimated state at time t , which is obtained as the weighted sum of all particles. Given a set of N particles $\{x_t^i\}_{i=1}^N$ with

their corresponding normalized weights $\{w_t^i\}_{i=1}^N$, the estimated state $\tilde{x}_t$ is computed as:

$$\hat{x}_t = \sum_{i=1}^N w_t^i x_t^i, \quad (23)$$

where w_t^i represents the normalized weight of the i th particle, ensuring that $\sum_{i=1}^N w_t^i = 1$. This weighted average provides an efficient estimate of the latent state while incorporating uncertainty from the particle distribution.

Using the particle filter approach, we obtain weekly estimates of the curvature parameter α in the probability weighting function. A lower(higher) value of α indicates a greater tendency to overweight(underweight) tail events. We follow [Shi et al. \(2023\)](#) and construct the Probability Weighting Index(PWI), defined as:

$$PWI = \frac{1}{\alpha}. \quad (24)$$

Since our estimation is conducted on a weekly basis, we compute the monthly average of the estimated PWI to represent the index value for each month in the subsequent empirical analysis on return predictability. In the next section, we describe several approaches to test the return predictability of the proposed predictor.

3.3 Return Predictability

3.3.1 In-Sample Test

The in-sample test evaluates the predictive power of our predictor within the estimation period by examining the statistical relationship between predictor variables and future stock returns. Specifically, we estimate the following predictive regression model:

$$r_{t+1} - r_{f,t+1} = \alpha + \beta X_t + \epsilon_{t+1}, \quad (25)$$

where r_{t+1} denotes the stock index return at time $t + 1$, and $r_{f,t+1}$ represents the risk-free rate at the same time. The parameter α is the intercept, while β captures the predictive relationship between the predictor X_t and excess return. Finally, ϵ_{t+1} is the error term.

To assess return predictability, we test the null hypothesis $H_0 : \beta = 0$, where rejection implies that the predictor variable contains significant information about the future returns. The regression is estimated using ordinary least squares (OLS), and statistical significance is evaluated using heteroskedasticity-robust standard errors to account for

potential non-constant variance in residuals. Model performance is further assessed using metrics such as the R^2 statistic, which measures the proportion of return variability explained by predictors.

In addition to evaluating the predictive power of our proposed predictor, we systematically compare its performance against a set of well-established predictors from the existing literature. Specifically, in all subsequent return predictability test, we include the 14 predictive variables used in [Goyal and Welch \(2008\)](#) as benchmarks. These variables, which have been extensively studied in the context of return predictability, allow us to assess the relative effectiveness of our predictor in comparison to traditional economic and financial indicators. A more detailed discussion of the predictors from [Goyal and Welch \(2008\)](#) will be provided in the empirical section.

Equation 3.23 represents a univariate predictive regression, where stock returns are regressed on a single predictor. To examine whether the predictive information embedded in PWI is distinct from that of our benchmark predictors, we extend the model to bivariate predictive regression by including both PWI and one of the [Goyal and Welch \(2008\)](#) predictors (GW predictors) as independent variables:

$$r_{t+1} - r_{f,t+1} = \alpha + \beta_{GW}X_{t,GW} + \beta_{PWI}X_{t,PWI} + \varepsilon_{t+1}, \quad (26)$$

where $X_{t,PWI}$ and $X_{t,GW}$ represent our proposed PWI predictor and one of the GW predictors, respectively. In this specification, a statistically significant estimate of β_{PWI} suggests that PWI retains predictive power for stock returns even after controlling for the information contained in traditional macroeconomic and financial variables. Conversely, an insignificant β_{PWI} would indicate that the predictive ability of PWI may be subsumed by existing predictors.

The bivariate predictive regression framework allows us to assess whether PWI provides incremental predictive value beyond widely studied predictors, thereby offering a more comprehensive evaluation of its role in return predictability.

3.3.2 Out-Sample Test

While strong in-sample performance is informative, it does not necessarily imply robust out-sample predictability. To rigorously assess the return predictability of our proposed predictor, we conduct two types of out-sample tests: the split-sample (one-time cut) approach and the expanding window approach. These methods allow us to evaluate the stability and practical applicability of our predictor in forecasting stock returns beyond the estimation period.

In the split-sample approach, we divide the entire sample into two subsets: an in-sample period and an out-of-sample period. We first estimate the predictive regression model in Equation 3.23 using data from the in-sample period and then use the estimated coefficients to forecast excess returns in the out-of-sample period. To evaluate the predictive performance of our proposed predictor, we employ the mean squared predictive error (MSPE) as a measure of forecast accuracy, defined as:

$$MSPE = \frac{1}{T - M} \sum_{t=M+1}^T (r_t - \hat{r}_t)^2, \quad (27)$$

where r_t is the realized monthly return, and $\hat{r}_t$ is the predicted return. Furthermore, we assess the relative predictive ability of our model using the out-of-sample R^2 statistic, calculated as:

$$R_{OOS}^2 = 1 - \frac{MSPE_{model}}{MSPE_{bench}}, \quad (28)$$

where $MSPE_{model}$ is the $MSPE$ of the model using our predictor, and $MSPE_{bench}$ is the $MSPE$ of a benchmark forecast. We adopt the historical average return as the benchmark prediction. Additionally, to provide a more comprehensive evaluation, we compare the out-of-sample R^2 of our *PWI* predictor against the 14 GW predictors, allowing us to assess its effectiveness in forecasting excess returns.

For the expansion window approach, we sequentially increase the size of the in-sample period as we move forward in time, re-estimating the predictive regression model at each step. Specifically, we begin with an initial estimation window ($n=36$) and expand it by adding new observations while keeping all past data, ensuring that each re-estimated model incorporates more information than the previous one. The estimated coefficients from each expanding window are then used to generate out-of-sample forecasts for the next period.

Formally, at each time t , we estimate the Equation 3.23 using all available data from an expanding window up to time t and then predict the excess return for time $t + 1$. This process continues until we reach the end of the sample. The predictive performance of the expanding window forecasts is evaluated using the $MSPE$ and out-of-sample R^2 metrics, as defined earlier.

Compared to the split-sample approach, the expanding window method allows for a more adaptive estimation, leveraging the increasing amount of historical information to improve parameter stability. This approach is particularly useful when structural breaks or time-varying relationships are present in the data, as it enables the model to adjust to evolving market conditions.

3.3.3 Economic Significance

In the previous section, we conducted statistical tests to evaluate the return predictability of the proposed predictor. However, statistical significance alone does not guarantee economic relevance. Even if a predictor demonstrates statistically significant predictive power, its actual economic impact—such as its effectiveness in portfolio allocation, trading strategies, or risk management—may be negligible. Therefore, to assess the practical value of our predictor, we further examine its economic significance.

Regarding the criterion for determining portfolio weights, we follow [Azimi et al. \(2024\)](#). Consider an expected utility-maximizing investor with constant relative risk aversion (CRRA). According to the investment and asset allocation model of [Merton \(1969\)](#), the optimal weight allocated to the aggregate market in the investor's portfolio is given by

$$w_t = \frac{1}{\lambda} \left[\frac{E_t[R_{t+1}] - R_{f,t}}{var_t[R_{t+1}]} \right], \quad (29)$$

where $E_t[R_{t+1}] - R_{f,t}$ represents the expected excess market return at time t , $var_t[R_{t+1}]$ denotes the conditional variance of the market return, and λ is the investor's coefficient of relative risk aversion. Specifically, w_t represents the proportion of wealth allocated to the market index, which, in our case, is the S&P 500 index. The remaining wealth is invested in the risk-free asset. This formulation implies that the optimal portfolio weight is positively related to the expected risk-adjusted return and inversely related to the investor's risk aversion.

To evaluate the performance of the investment strategy based on our proposed predictor, we employ the certainty equivalent return (CEQ) metric, which represents the risk-free rate an investor would be willing to accept in lieu of holding the risky portfolio. For a mean-variance investor with a risk aversion parameter γ , the CEQ is defined as:

$$CEQ = E(R_P) - \frac{\gamma}{2} Var(R_P), \quad (30)$$

where $E(R_P)$ and $Var(R_P)$ denote the mean and variance of the portfolio return, respectively, computed over the out-of-sample evaluation period. Although our model is derived under CRRA utility, we consider the CEQ metric to be a valid measure of investment performance, as it effectively quantifies the trade-off between expected return and risk, aligning with standard portfolio evaluation methodologies.

To implement the portfolio allocation framework described above, we must specify the inputs for Equation 3.27. Since risk aversion is known to vary over time, we consider three different levels, setting $\lambda = 1, 2, 3$ and take the average CEQ across these values to account

for the time-varying nature of risk preferences.

Consistent with the expanding window approach introduced earlier, we re-estimate the predictive regression model at each time t and use the resulting estimates to forecast the return at time $t + 1$, which serves as $E_t[R_{t+1}]$ in Equation 3.27. In addition to the univariate predictive regression, we also estimate a bivariate predictive regression, incorporating both our proposed *PWI* predictor and one of the GW predictors. This allows us to assess whether the inclusion of *PWI* enhances investment performance by comparing the resulting CEQ to that obtained using only a GW predictor. For the conditional variance $Var_t[R_{t+1}]$, we adopt a simple approach by computing the realized return variance over the previous month, which serves as a proxy for investors' expectations of future volatility.

To ensure realistic portfolio constraints, we impose bounds on the portfolio weight w_t , restricting it to the range $[-0.5, 1.5]$. This prevents excessively large short positions and leverage, making the allocation strategy more practical for real-world implementation.

3.3.4 Forecast Combination

In addition to univariate and bivariate predictive regressions, we extend our analysis to multivariate approaches, such as forecast combination. [Rapach et al. \(2010\)](#) show that combining forecasts from multiple univariate models can yield reliable return predictability, even when no individual model significantly outperforms the historical average benchmark. Building on this insight, we compare the CEQ under two different forecast combinations: one using the 14 GW predictors alone and the other incorporating both the 14 GW predictors and our proposed *PWI* predictor. This comparison allows us to assess whether the inclusion of *PWI* enhances investment performance. Mathematically, the combined forecast is computed as:

$$\hat{r}_{t,combine} = \sum_{i=1}^N w_{i,t} \hat{r}_{i,t}, \quad (31)$$

where $\hat{r}_{i,t}$ represents the return forecast produced by the i th predictor, $w_{i,t}$ is the weight assigned to the i th predictor, and $\hat{r}_{t,combine}$ is the aggregate forecast.

Following [Wang et al. \(2019\)](#), we employ multiple weighting schemes to determine $w_{i,t}$, ensuring robustness across different combination methods. We introduce these weighting approaches in detail in the following.

The first approach we employ is the equal-weighted combination (EWC), which assigns an equal weight to all predictors. This method serves as a simple benchmark by averaging forecasts without favoring any particular predictor.

The second approach, the trimmed mean combination (TMC), is a slight modification of EWC. In TMC, we exclude the predictors with the worst predictive performance in the previous period, as measured by MSPE. By removing underperforming predictors, TMC aims to enhance forecast accuracy while maintaining a simple averaging structure.

The third and fourth approaches introduce a performance-based weighting scheme, known as discounted MSPE (DMSPE), which was proposed by [Stock and Watson \(2004\)](#). Under this scheme, the weight assigned to the i th predictor is given by

$$w_{i,t} = \frac{\Phi_{i,t-1}^{-1}}{\sum_{j=1}^N \Phi_{j,t-1}^{-1}}, \quad (32)$$

where $\Phi_{i,t} = \sum_{k=1}^t \delta^{t-k} (r_k - \hat{r}_{i,k})^2$. This formulation ensures that more recent forecast errors receive greater weight, allowing the model to adapt to changing market conditions. The difference between the third and fourth approaches lies in the choice of the discount factor δ : we set $\delta = 1$ in the third approach, implying equal weighting of past errors, while in the fourth approach, we set $\delta = 0.9$, placing greater emphasis on recent performance.

The final approach, proposed by [Yang \(2014\)](#), adopts a nonlinear weighting scheme that applies an exponential transformation of MSPE to assign weights dynamically. The weight for the i th predictor is given by:

$$w_{i,t} = \frac{\pi_i \exp(-\lambda \sum_{k=1}^{t-1} (r_k - \hat{r}_{i,k})^2)}{\sum_{j=1}^N \pi_j \exp(-\lambda \sum_{k=1}^{t-1} (r_k - \hat{r}_{j,k})^2)}. \quad (33)$$

Following the standard practice in the literature, we set $\pi_i = 1$ and $\lambda = 1$. This approach ensures that predictors with lower past MSPE receive exponentially higher weights, making it particularly effective in filtering out poor predictors while dynamically adapting to evolving predictive performance.

By employing these different weighting schemes, we aim to provide a comprehensive evaluation of whether combining forecasts—particularly with the inclusion of *PWI*—enhances investment performance.

3.4 Predict Channel

Classical valuation models suggest that stock prices are driven by both expected future cash flows and the discount rates applied to them. Mathematically, the log-linearization

of stock returns by [Campbell and Shiller \(1988\)](#) yields the following approximate identity:

$$R_{t+1} = k + DG_{t+1} - \rho D/P_{t+1} + D/P_t, \quad (34)$$

where R_{t+1} represents the aggregate stock market return from time t to $t+1$, DG_{t+1} denotes the log aggregate dividend growth rate, D/P_{t+1} is the log aggregate dividend-price ratio, and ρ is a positive log-linearization constant.

Equation 3.32 implies that if a variable can predict stock returns beyond the information contained in D/P_{t+1} , it must do so by forecasting either future dividend growth (DG_{t+1}) or the future dividend-price ratio (D/P_{t+1}), or both. Based on this insight, we investigate whether our proposed *PWI* predictor forecasts stock returns through the expected cash flow channel, the discount rate channel, or both, by estimating the following predictive regression:

$$Y_{t+1} = \alpha + \beta \cdot PWI_t + \phi D/P_t + \epsilon_t, \quad (35)$$

where Y_{t+1} represents a proxy for either expected cash flows or discount rates at time $t+1$, PWI_t is our proposed predictor, and D/P_t is the dividend-price ratio at time t , included to control for its well-documented role in return predictability.

Following [Wang et al. \(2019\)](#), we employ different proxies for Y_{t+1} to separately identify the predictive channels. Specifically, for the discount rate channel, we use the dividend-price ratio (D/P) and long-term bond yield (LTY) as proxies, as they reflect variations in required returns. For the expected cash flow channel, we use dividend growth (DG) and earnings growth (EG) as proxies, which capture fluctuations in future cash flow expectations.

By examining the statistical significance of β across these different specifications, we assess whether *PWI* primarily predicts stock returns by capturing variations in expected cash flows, discount rates, or a combination of both.

4 Data

4.1 Option, index, and risk-free rate

Based on the previous discussion, our empirical analysis requires data on S&P 500 index options, the S&P 500 index, and the risk-free rate. The S&P 500 index option data is used to estimate the curvature parameter α in the probability weighting function, while the S&P 500 index returns and the risk-free rate are utilized in the return predictability tests.

We obtain S&P 500 index option data from OptionMetrics, S&P 500 index data from

CRSP, and the risk-free rate from the FRED database, where we use the 3-month Treasury bill rate as the proxy for the risk-free rate in our empirical analysis. Our sample period spans January 2010 to August 2023, and we apply a series of data filters to the option data to ensure its reliability for analysis.

Following [Rosenberg and Engle \(2002\)](#) and [Polkovnichenko and Zhao \(2013\)](#), we select option data from Wednesdays to maintain consistency and reduce potential day-of-the-week effects. We include call options with deltas ranging from 0.1 to 0.55 and put options with deltas ranging from -0.55 to -0.11, ensuring that our estimation primarily relies on at-the-money (ATM) and out-of-the-money (OTM) options, which are generally more liquid and informative for extracting risk-neutral probabilities.

To ensure data quality and liquidity, we impose several additional screening criteria. We exclude options with zero trading volume or zero option interest that the mid-quote price exceeds 3/8. Additionally, the annualized implied volatility must lie within a reasonable range of 5% to 90% to filter out extreme pricing anomalies.

Furthermore, to enforce no-arbitrage conditions, we discard options that violate the lower bound constraint, given by:

$$P_t \geq \max[0, \exp[-r_f(T-t)K - S_t + D_{t,T}]] \quad \text{or} \quad C_t \geq \max[0, S_t - \exp[-r_f(T-t)K - D_{t,T}]], \quad (36)$$

where P_t and C_t denote put and call option prices, respectively, S_t is the underlying asset price, K is the strike price, r_f is the risk-free rate, and $D_{t,T}$ represents dividends paid before expiration. Additionally, we ensure that option prices satisfy the maximum vertical premium condition, such that

$$C_t(K_1) - C_t(K_2) \leq K_2 - K_1, \quad P_t(K_2) - P_t(K_1) \leq K_2 - K_1. \quad (37)$$

Any options that fail to meet these criteria are excluded from our dataset to maintain the integrity and reliability of our empirical analysis.

After applying the initial screening criteria, a large number of option contracts remain available on each trading day. This significantly increases the computational burden of our particle filter approach. To address this issue, we impose additional refinements to further streamline the dataset while preserving its informational content.

First, we select options with a time to maturity closest to 30 days, ensuring that we focus on contracts with consistent time horizons across estimation periods. Following [Li \(2020\)](#), we categorize the remaining options into five moneyness intervals and select one representative option from each interval. Specifically, we choose the option whose mon-

eyness is closest to the midpoint of each interval. This method ensures that our dataset includes options across different strike price levels, providing a balanced representation for estimation, while simultaneously preventing an excessive number of contracts that could lead to computational inefficiency.

[Insert Table 1 here]

Table 1 presents summary statistics for the refined option dataset, categorized by moneyness intervals, which is defined as $\log(K/S)$. The dataset consists of 6990 option contracts, with implied volatility (IV) ranging from 5.5% to 89.7%. The average implied volatility varies across intervals, with deeper out-of-the-money (OTM) options generally exhibiting higher IV levels.

Trading activity also differs by moneyness, as reflected in the average trading volume and open interest. The highest mean trading volume is observed in the near-the-money interval ($[-0.03, 0.00]$), with an average of 3011 contracts per day, suggesting greater liquidity in this range. In contrast, deep OTM options have significantly lower trading volume and open interest, consistent with the market preference for more liquid contracts.

These descriptive statistics confirm that the refined dataset maintains a diverse representation across different strike price levels while ensuring a manageable number of contracts for empirical analysis.

4.2 Candidate predictors

To evaluate the effectiveness and predictive power of our proposed predictor and to situate our findings within the broader literature on stock return predictability, we incorporate the 14 predictive variables introduced by Goyal and Welch (2008). These variables, which have been widely studied in the forecasting literature, serve as benchmarks for comparison. The dataset can be obtained from Amit Goyal’s website, where the authors have made the data publicly available.

[Insert Table 2 here]

Table 2 provides a detailed overview of these 14 GW predictors, including their full names, abbreviations, and definitions. These variables capture a range of economic, financial, and market-related factors that have been shown to exhibit predictive power in prior studies.

Furthermore, Table 3 presents the summary statistics of our proposed *PWI* predictor alongside the GW predictors. Specifically, the table reports the mean, standard deviation,

minimum, median, and maximum values for each variable, providing a comprehensive overview of their distributional characteristics.

[Insert Table 3 here]

5 Empirical Results

5.1 Estimation of the curvature

In section 3.2.2, we introduced three volatility models—ARCH, GARCH, and EGARCH—to fit the stock return data in our sample. Table 4 presents the estimation results. Among the three models, EGARCH exhibits the highest log-likelihood value and the lowest Akaike Information Criterion (AIC), suggesting that it provides the best fit for the data.

[Insert Table 4 here]

Additionally, we find that both the ζ (fat-tail parameter) and κ (asymmetry parameter) estimates are statistically significant. This indicates that the stock return distribution exhibits both fat tails and skewness, consistent with well-documented empirical characteristics of financial returns. Based on these findings, we adopt the EGARCH model to simulate the stock return distribution in our subsequent analysis.

As an example, we examine the EGARCH-simulated stock return distribution for March 4, 2020. Figure 3 presents the simulated distribution, generated using the past 1250 daily returns as input for the volatility model. As described earlier, we apply kernel density estimation (KDE) with a bandwidth of $0.9N^{-1/5}\sigma$ to smooth the distribution and discretize it into 200 grid points, reflecting the assumption of a finite state space under our model.

The simulated return distribution exhibits significant fat tails and skewness, consistent with the empirical characteristics of stock returns. This suggests that the EGARCH model effectively captures asymmetric volatility dynamics, reinforcing its suitability for modeling real-world return behavior.

[Insert Figure 3 here]

Once the physical return distribution is estimated, we can price options using Equation 3.8 and extract the curvature parameter α by integrating the empirical pricing kernel approach with the particle filter methodology.

Figure 4 represents the estimation results for the curvature parameter in the probability weighting function (PWF). To facilitate interpretation, we transform α into the probability

weighting index (*PWI*) by taking its reciprocal and then compute the monthly average. In the figure, the line with x markers corresponds to the monthly average of $1/\alpha$, which we define as *PWI*.

[Insert Figure 4 here]

In the context of the probability weighting function, a curvature parameter $\alpha < 1$ implies that investors overweight tail events, assigning excessive subjective probabilities to extreme outcomes. Conversely, when $\alpha > 1$, investors underweight tail events, perceiving extreme events as less likely than their true probability suggest. The behavioral finance literature generally finds that α tends to be less than 1, indicating a persistent overweighting of tail events, with experimental calibrations often yielding values around $\alpha = 0.65$.

However, a growing body of research emphasizes that the curvature parameter of the probability weighting function is time-varying and does not always exhibit a constant overweighting of tail events (Polkovnichenko and Zhao, 2013; Shi et al., 2023). Our estimation results confirm this dynamic nature, showing that the shape of the probability weighting function evolves over time. Notably, during the COVID-19 pandemic period (February 2020-April 2020), we observe a sharp increase in *PWI* (corresponding to a decrease in α), suggesting that investors assigned greater weight to tail events, which is consistent with heightened risk perception and uncertainty in financial markets.

Given these findings, we further investigate whether *PWI* serves as a predictor of aggregate stock returns, as it reflects shifts in investor psychology and may exert an indirect influence on market dynamics.

5.2 In-sample results

To assess the predictive power of our proposed predictor, we first estimate the univariate predictive regression over the full sample period, as outlined in Section 3.3.1. Table 5 presents the estimation results for Equation 3.23, where all predictors are standardized.

Several key findings emerge from the analysis. First, among the 15 predictors, only five exhibit statistical significance, one of which is our proposed *PWI*. Second, the coefficient of *PWI* is positive, indicating that a higher degree of investor overweighting of tail events is associated with higher expected stock returns. Specifically, a one-standard-deviation increase in this month's *PWI* is associated with a 0.0058 increase in next month's excess return. Third, the predictive coefficient of *PWI* is larger than that of most GW predictors, suggesting that it has relatively stronger predictive power. Lastly, the R^2 of *PWI* exceeds that of 11 out of the 14 GW predictors, further highlighting its effectiveness in explaining

return variations. Overall, these results suggest that *PWI* performs well in the univariate predictive regression, providing meaningful insights into return predictability.

[Insert Table 5 here]

In addition to the univariate predictive regression, we also estimate the bivariate predictive regression, where each GW predictor is included as a control variable. Table 6 presents the estimation results for Equation 3.24.

Several notable findings emerge. First, the predictive coefficient of *PWI* remains statistically significant in 9 out of 14 cases, indicating that its predictive power persists even after controlling for GW predictors. Second, the R^2 of each bivariate model is consistently higher than that of the corresponding univariate model, suggesting that the inclusion of *PWI* enhances explanatory power. Third, the estimated predictive coefficients of *PWI* remain relatively stable across different specifications, ranging between 0.0024 and 0.009, implying robustness to the choice of control variable.

[Insert Table 6 here]

These results indicate that *PWI* provides distinct predictive information beyond what is captured by GW predictors. The fact that *PWI* remains significant even after controlling for traditional predictors, coupled with the improvement in R^2 , suggests that it captures an additional dimension of return predictability, potentially linked to investor's risk perception.

5.3 Out-of-sample results

In this section, we examine the out-of-sample performance of *PWI*, as strong in-sample predictive ability does not necessarily translate into robust out-of-sample performance, as previously discussed. Table 7 presents the estimation results obtained using the split-sample approach described in Section 3.3.2.

The results show that in the out-of-sample period, our *PWI* predictor achieves the highest R^2 of 2.18%, whereas only 4 out of the 14 GW predictors exhibit a positive R^2 . Furthermore, the average out-of-sample R^2 across all 14 GW predictors is -5.24%, indicating that most of these predictors fail to provide meaningful forecasts.

[Insert Table 7 here]

These findings highlight that our *PWI* predictor far surpasses the alternative predictors in out-of-sample predictive accuracy. The *PWI* predictor's mean squared error (MSPE) is

lower than the benchmark, whereas the MSPEs of the candidate exceed the benchmark, further reinforcing its superior predictive performance.

To further assess the out-of-sample performance of *PWI*, we extend our analysis by employing the expanding window approach. Unlike the split-sample method, which uses a fixed in-sample period, this approach gradually increases the estimation window over time, incorporating more information as new data becomes available. This allows for a more dynamic evaluation of the predictor's forecast ability.

[Insert Table 8 here]

Table 8 presents the out-of-sample R^2 results using this method. The first column reports the one-month-ahead forecast results. Consistent with prior literature (Goyal and Welch, 2008; Rapach et al., 2010), most of the predictors can not outperform the historical average benchmark, as indicated by their negative R^2 values. This aligns with the well-documented finding that the historical average is difficult to surpass in out-of-sample forecasting. However, among all predictors, *PWI* exhibits R^2 of -1.16% which is higher than most of the GW predictors except NTIS and DFY, suggesting that its predictive performance is closer to the historical average than most of the competing candidate predictors.

Beyond the one-month horizon, we also examine longer forecasting horizons, including two-month, three-month, and six-month periods. As shown in Table 4.8, the out-of-sample R^2 of *PWI* declines as the forecasting horizon lengthens, indicating that *PWI* is more effective for short-term return prediction. Overall, while none of the predictors outperform the benchmark, *PWI* consistently achieves higher R^2 values than the average GW predictor across all horizons, reinforcing its relative strength in stock return forecasting.

5.4 Economic significance

In this section, we evaluate the economic significance of our proposed *PWI* index, by constructing an investment strategy. Specifically, we consider an investor who dynamically allocates wealth between a risky asset (the market portfolio) and a risk-free asset based on the predictive signals from *PWI*. The investor determines the optimal weight ex-ante using Equation 3.27, aiming to maximize expected utility while accounting for risk.

Table 9 reports the CEQ for the buy-and-hold strategy and the 15 candidate predictors, including our proposed *PWI*. Using the univariate predictive regression, we obtain an average CEQ gain of 7.74% across different levels of risk aversion, highlighting the economic significance of incorporating *PWI* in portfolio allocation. In comparison, the buy-and-hold benchmark yields an average CEQ gain of 6.62%, indicating that *PWI*-based strategies provide superior risk-adjusted returns.

[Insert Table 9 here]

Furthermore, *PWI* outperforms nearly all GW predictors, with the exception of LOG_DP and LOG_DY, which have been recognized in prior literature as strong predictors of stock returns. Additionally, *PWI*'s CEQ exceeds the average CEQ of GW predictors across different levels of risk aversion, as shown in the last row of Table 9.

[Insert Table 10 here]

In the previous section, we demonstrated that *PWI* contains incremental predictive information beyond that captured by the GW predictors, as evidenced by the bivariate predictive regression results. This finding suggests that *PWI* and the GW predictors convey distinct signals regarding future stock returns. Building on this insight, we assess whether *PWI* as an additional predictive variable enhances investment performance. Specifically, for each GW predictor, we compute the CEQ from a bivariate predictive regression that includes both *PWI* and the respective GW predictor. We then compare the CEQ obtained from the bivariate model to that from the corresponding univariate model, evaluating whether the inclusion of *PWI* leads to improved economic performance.

Table 10 presents the CEQ gains for portfolios constructed using predictions from the bivariate predictive model. The results indicate that adding *PWI* as an additional predictive variable improves investment performance in half of the cases. For instance, the average CEQ of the BM-based model increases by 0.47 percentage points, rising from 2.86% to 3.33%, while the LOG_EP-based model experiences an increase of 0.65 percentage points, from 1.7% to 2.35%.

These findings, along with the results from the univariate and bivariate predictive models, suggest that *PWI* provides significant economic value in asset allocation decisions and performs competitively against well-established return predictors. To further evaluate the effectiveness of *PWI* and strengthen our analysis, we employ the forecast combination approach, which integrates predictions from multiple predictors by assigning weights on each. The methodology for determining these weights has been detailed in Section 3.3.4.

Using the combined forecast along with Equation 3.27, we determine the optimal portfolio allocation between the risky asset and the risk-free asset, following the same methodology applied in the univariate and bivariate predictive models. Table 11 presents the results for the five forecast combination approaches. Consistent with prior literature, all five methods yield similar investment performance, suggesting robustness across different weighting schemes. Notably, incorporating *PWI* into the forecast combinations enhances investment performance across all five approaches, as measured by the average

CEQ across varying level of risk aversion. This improvement further strengthen the notion that *PWI* captures additional predictive information beyond the GW predictors. Specifically, the inclusion of *PWI* results in an average CEQ increase of 42.5 basis points, highlighting the potential contribution to portfolio decision-making.

[Insert Table 11 here]

This section examines the economic significance of *PWI* in portfolio allocation, we find that *PWI*-based strategies yield higher CEQ gains than the buy-and-hold benchmark and most GW predictors. Adding *PWI* in bivariate regressions improves investment performance in several cases, and incorporating it into forecast combinations enhances CEQ across all weighting schemes. These findings confirm that *PWI* provides valuable return-predictive information beyond traditional predictors.

5.5 Channel of predictability

We have established the return predictability of *PWI*, but understanding the underlying mechanism driving this predictability is equally important. In this section, we investigate the sources of *PWI*'s predictive power at the market level, offering insights into how it forecasts stock returns.

Table 12 presents the estimation results of Equation 3.33, revealing distinct patterns in the predictive relationship between *PWI* and proxies for cash-flow and discount-rate dynamics. The coefficients of *PWI* for DP and LTY are -0.0308 and -0.8019, respectively, both statistically significant at the 1% level. These results indicate that *PWI* exhibits strong predictive power for discount-rate proxies. The negative coefficients imply a positive relationship between *PWI* and expected excess market returns, which is consistent with the return predictability results reported in Table 5. In contrast, the coefficient of *PWI* for DG is -0.0389 and statistically insignificant, while for EG, it is -0.0872 and significant at the 10% level. However, the negative coefficients contradict the expected relationship between *PWI* and future returns, as a negative association with cash-flow proxies would suggest a decline in expected market returns, inconsistent with our previous findings.

[Insert Table 12 here]

Overall, the strong negative predictability of *PWI* for DP and LTY, coupled with its weak relationship with DG and EG, suggests that the return predictability of *PWI* primarily operates through the discount-rate channel. This aligns with the time-varying

discount-rate framework widely documented in the literature. Specifically, a higher *PWI*—which reflects increased probability weighting on extreme events—may indicate heightened risk perceptions among investors. As a result, investors demand higher risk premia, leading to a lower discount rate and higher expected returns. This interpretation supports the notion that *PWI* captures shifts in risk preferences, influencing asset prices through changes in required returns [Polkovnichenko and Zhao \(2013\)](#).

6 Conclusion

Traditional stock return predictability research has primarily focused on macro predictors, often regarded as “rational” predictors. However, with the rise of behavioral finance and the increasing presence of retail investors in the market, researchers have begun exploring “irrational” predictors, the most common being investor sentiment. While sentiment captures general market mood, optimism, or pessimism, it fails to account for underlying risk perception distortions driven by cognitive biases.

Building on this insight, we investigate whether the curvature parameter of the probability weighting function—which reflects investors’ perception of risk within a behavioral finance framework—exhibits return predictability and whether an investment strategy based on probability weighting can outperform a buy-and-hold strategy. To achieve this, we employ a particle filter approach combined with an empirical pricing kernel approach to obtain a smooth, dynamic estimation of the curvature parameter, which we then transform into the Probability Weighting Index (*PWI*). We conduct both in-sample and out-of-sample tests to evaluate its predictive power, and the results indicate that *PWI* exhibits stronger return predictability than most of the popular predictors. Additionally, we construct an investment strategy based on *PWI* and demonstrate its economic significance through portfolio performance improvements. Finally, we analyze the underlying mechanism driving *PWI*’s predictive power, finding that it primarily operates through the discount rate channel.

By conducting a comprehensive analysis of *PWI*’s return predictability, we demonstrate that it serves as a potential “irrational” predictor of market returns, offering a new perspective on the role of cognitive biases in asset pricing. Our findings take the understanding of how probability distortions influence market dynamics one step further, bridging the gap between behavioral finance and return predictability. Moreover, by establishing a direct link between probability weighting and investment decision-making, our research provides a foundation for future studies exploring the implications of cognitive biases in financial markets.

References

- Azimi, M., S. Ghazi, and M. Schneider (2024). Probability weighting and equity premium prediction: Investing with optimism. *Financial Management*.
- Baele, L., J. Driessen, S. Ebert, J. M. Londono, and O. G. Spalt (2019). Cumulative prospect theory, option returns, and the variance premium. *The Review of Financial Studies* 32(9), 3667–3723.
- Barber, B. M. and T. Odean (2000). Trading is hazardous to your wealth: The common stock investment performance of individual investors. *The Journal of Finance* 55(2), 773–806.
- Barberis, N. and M. Huang (2008). Stocks as lotteries: The implications of probability weighting for security prices. *American Economic Review* 98(5), 2066–2100.
- Boudoukh, J., R. Michaely, M. Richardson, and M. R. Roberts (2007). On the importance of measuring payout yield: Implications for empirical asset pricing. *The Journal of Finance* 62(2), 877–915.
- Breeden, D. T. and R. H. Litzenberger (1978). Prices of state-contingent claims implicit in option prices. *Journal of Business*, 621–651.
- Brown, G. W. and M. T. Cliff (2005). Investor sentiment and asset valuation. *The Journal of Business* 78(2), 405–440.
- Campbell, J. Y. (1987). Stock returns and the term structure. *Journal of Financial Economics* 18(2), 373–399.
- Campbell, J. Y. and R. J. Shiller (1988). The dividend-price ratio and expectations of future dividends and discount factors. *The Review of Financial Studies* 1(3), 195–228.
- Campbell, J. Y. and S. B. Thompson (2008). Predicting excess stock returns out of sample: Can anything beat the historical average? *The Review of Financial Studies* 21(4), 1509–1531.
- Charupat, N., R. Deaves, T. Derouin, M. Klotzle, and P. Miu (2013). Emotional balance and probability weighting. *Theory and Decision* 75, 17–41.
- Chateauneuf, A., J. Eichberger, and S. Grant (2007). Choice under uncertainty with the best and worst in mind: Neo-additive capacities. *Journal of Economic Theory* 137(1), 538–567.

- Chen, T.-Y., Y.-L. Lin, and L. Y. Tzeng (2024). Estimating probability weighting functions through option pricing bounds. *The Review of Asset Pricing Studies* 14(3), 513–543.
- Chiang, I.-H. E. and W. K. Hughen (2017). Do oil futures prices predict stock returns? *Journal of Banking & Finance* 79, 129–141.
- Cochrane, J. H. (2008). The dog that did not bark: A defense of return predictability. *The Review of Financial Studies* 21(4), 1533–1575.
- De Giorgi, E. G. and S. Legg (2012). Dynamic portfolio choice and asset pricing with narrow framing and probability weighting. *Journal of Economic Dynamics and Control* 36(7), 951–972.
- Driessen, J., S. Ebert, and J. Koëter (2025). π -capm: The classical capm with probability weighting and skewed assets. *Review of Financial Studies*.
- Fama, E. F. and K. R. French (1989). Business conditions and expected returns on stocks and bonds. *Journal of Financial Economics* 25(1), 23–49.
- Fama, E. F. and G. W. Schwert (1977). Asset returns and inflation. *Journal of Financial Economics* 5(2), 115–146.
- Gonzalez, R. and G. Wu (1999). On the shape of the probability weighting function. *Cognitive Psychology* 38(1), 129–166.
- Goyal, A. and I. Welch (2008). A comprehensive look at the empirical performance of equity premium prediction. *The Review of Financial Studies* 21(4), 1455–1508.
- He, X. D. and Y. Sun (2023). Dynamic portfolio selection and asset pricing under neo-additive probability weighting. *Available at SSRN 4455110*.
- Huang, D., F. Jiang, J. Tu, and G. Zhou (2015). Investor sentiment aligned: A powerful predictor of stock returns. *The Review of Financial Studies* 28(3), 791–837.
- Kumar, A. (2009). Hard-to-value stocks, behavioral biases, and informed trading. *Journal of Financial and Quantitative Analysis* 44(6), 1375–1401.
- Lewellen, J. (2004). Predicting returns with financial ratios. *Journal of Financial Economics* 74(2), 209–235.
- Li, B. (2020). Option-implied filtering: evidence from the garch option pricing model. *Review of Quantitative Finance and Accounting* 54(3), 1037–1057.

- Lopes, H. F. and R. S. Tsay (2011). Particle filters and bayesian inference in financial econometrics. *Journal of Forecasting* 30(1), 168–209.
- Ludvigson, S. C. and S. Ng (2007). The empirical risk–return relation: A factor analysis approach. *Journal of Financial Economics* 83(1), 171–222.
- McLean, R. D. and J. Pontiff (2016). Does academic research destroy stock return predictability? *The Journal of Finance* 71(1), 5–32.
- Meng, J. and X. Weng (2018). Can prospect theory explain the disposition effect? a new perspective on reference points. *Management Science* 64(7), 3331–3351.
- Merton, R. C. (1969). Lifetime portfolio selection under uncertainty: The continuous-time case. *The Review of Economics and Statistics*, 247–257.
- Polkovnichenko, V. and F. Zhao (2013). Probability weighting functions implied in options prices. *Journal of Financial Economics* 107(3), 580–609.
- Prelec, D. (1998). The probability weighting function. *Econometrica*, 497–527.
- Quiggin, J. (1982). A theory of anticipated utility. *Journal of Economic Behavior & Organization* 3(4), 323–343.
- Rapach, D. E., J. K. Strauss, and G. Zhou (2010). Out-of-sample equity premium prediction: Combination forecasts and links to the real economy. *The Review of Financial Studies* 23(2), 821–862.
- Rosenberg, J. V. and R. F. Engle (2002). Empirical pricing kernels. *Journal of Financial Economics* 64(3), 341–372.
- Shefrin, H. and M. Statman (1985). The disposition to sell winners too early and ride losers too long: Theory and evidence. *The Journal of finance* 40(3), 777–790.
- Shi, Y., X. Cui, and X. Y. Zhou (2023). Beta and coskewness pricing: Perspective from probability weighting. *Operations Research* 71(2), 776–790.
- Shleifer, A. and R. W. Vishny (1997). The limits of arbitrage. *The Journal of Finance* 52(1), 35–55.
- Stambaugh, R. F. (1999). Predictive regressions. *Journal of Financial Economics* 54(3), 375–421.

- Stock, J. H. and M. W. Watson (2004). Combination forecasts of output growth in a seven-country data set. *Journal of Forecasting* 23(6), 405–430.
- Tversky, A. and D. Kahneman (1992). Advances in prospect theory: Cumulative representation of uncertainty. *Journal of Risk and Uncertainty* 5, 297–323.
- Wang, Y., Z. Pan, L. Liu, and C. Wu (2019). Oil price increases and the predictability of equity premium. *Journal of Banking & Finance* 102, 43–58.

Figures

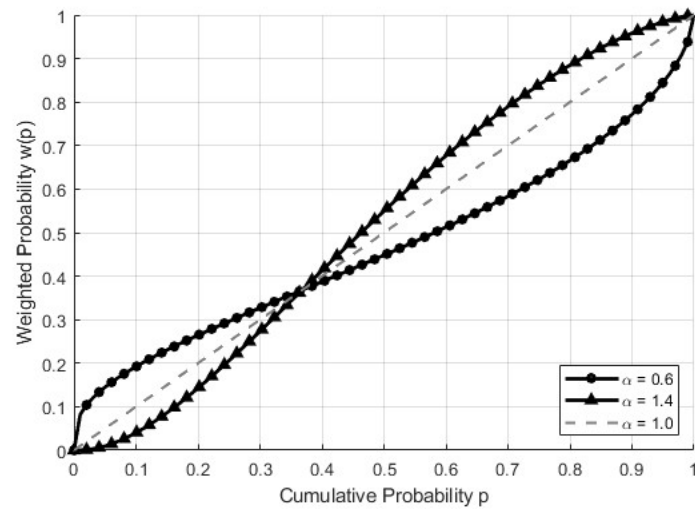


Figure 1: Prelec's probability weighting function with different levels of α .

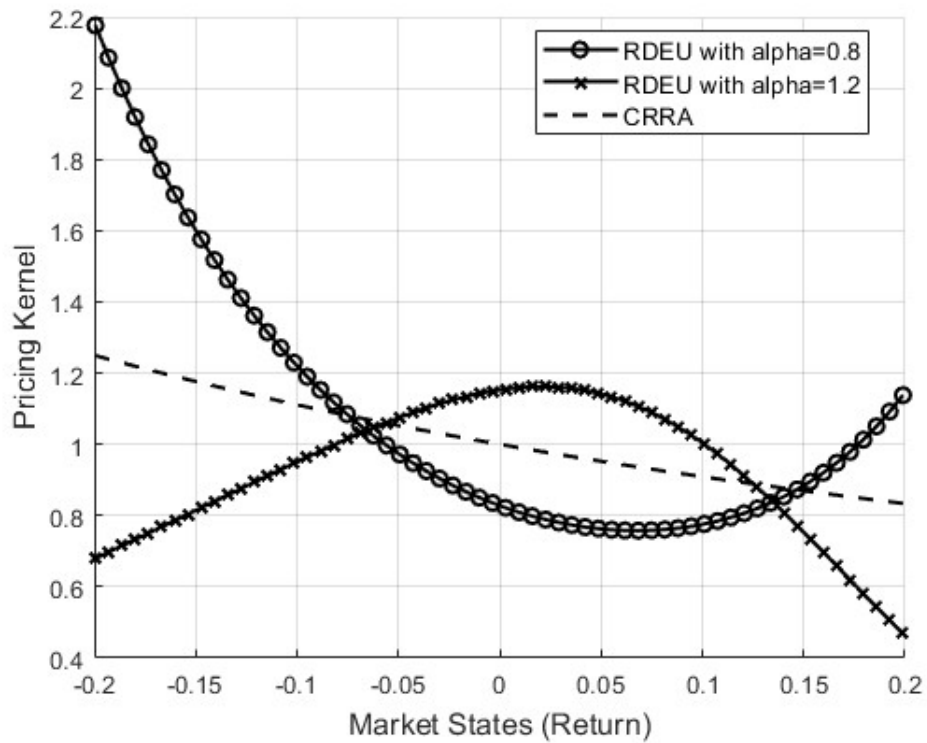


Figure 2: Theoretical pricing kernel.

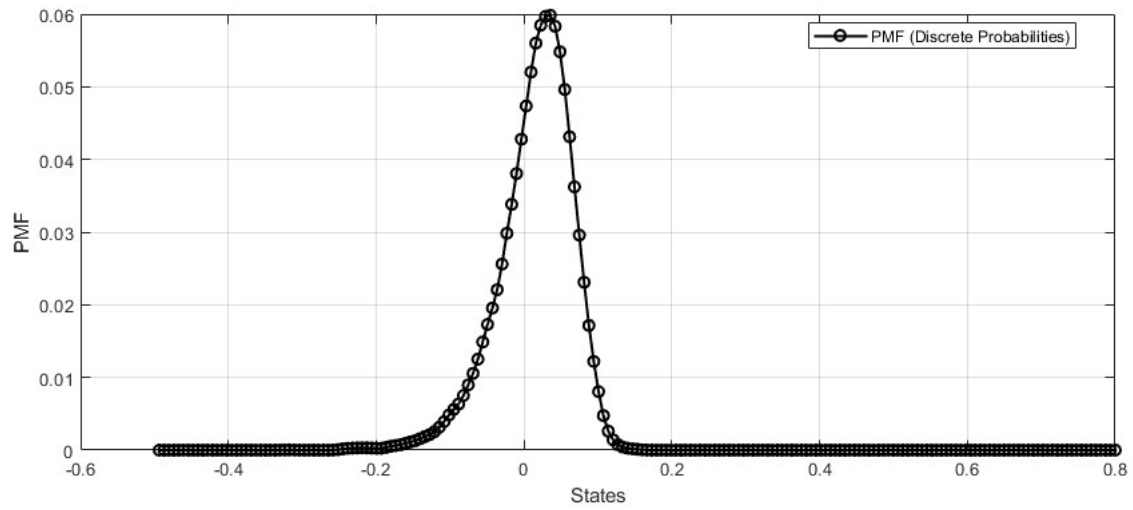


Figure 3: Estimated S&P 500 return density functions on March 4, 2020.

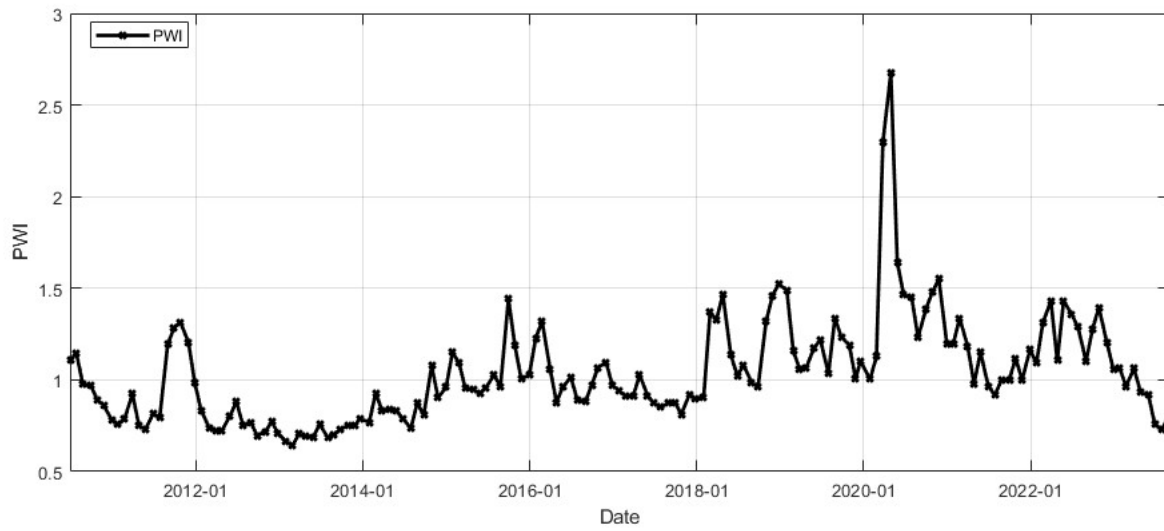


Figure 4: Monthly probability weighting index.

Tables

Table 1: Summary of option data used for pricing kernel estimation

Interval	Count	IV_Mean	IV_Max	IV_Min	Vol_Mean	OI_Mean
$[-\infty, -0.05]$	1125	0.266	0.897	0.132	1508	13474
$[-0.05, -0.03]$	833	0.195	0.730	0.102	1888	16395
$[-0.03, 0.00]$	1558	0.156	0.633	0.064	3011	14981
$[0.00, 0.03]$	2173	0.149	0.730	0.055	2543	13458
$[0.03, 0.05]$	953	0.143	0.582	0.061	1635	12025
$[0.05, \infty]$	348	0.217	0.663	0.100	1014	9100
Overall	6990	0.177	0.897	0.055	2203	13738

Note: The data used in this study is sourced from OptionMetrics via WRDS, covering the period from 2010 to 2023. Each Wednesday, we extract a cross-section of at-the-money (ATM) and out-of-the-money (OTM) options, including both puts and calls, with approximately one month to expiration. To ensure the reliability and representativeness of the extracted data, we apply a series of data screening criteria that filter out potential inconsistencies and market anomalies. Our final sample consists of 685 estimation dates and a total of 6,990 options, providing a robust dataset for analysis.

Table 2: Candidate Predictors for Market Excess Return Forecasting

GW Predictors (2008)	Abbreviation	Definition
Dividend-price ratio	LOG_DP	Difference between the log of dividends paid on the S&P 500 index and the log of stock prices (S&P 500 index), measured using a one-year moving sum.
Dividend yield (log)	LOG_DY	Difference between the log of dividends and the log of lagged stock prices.
Earnings-price ratio (log)	LOG_EP	Difference between the log of earnings on the S&P 500 index and the log of stock prices, where earnings are measured using a one-year moving sum.
Dividend-payout ratio (log)	LOG_DE	Difference between the log of dividends and the log of earnings.
Stock variance	SVAR	Sum of squared daily returns on the S&P 500 index.
Book-to-market ratio	BM	Ratio of book value to market value for the Dow Jones Industrial Average.
Net equity expansion	NTIS	Ratio of twelve-month moving sums of net issues by NYSE-listed stocks to total end-of-year market capitalization of NYSE stocks.
Treasury bill rate	TBL	Interest rate on a three-month Treasury bill (secondary market).
Long-term yield	LTY	Yield on long-term government bonds.
Long-term return	LTR	Return on long-term government bonds.
Default yield spread	DFY	Difference between BAA- and AAA-rated corporate bond yields.
Default return spread	DFR	Difference between the return on long-term corporate bonds and long-term government bonds.
Inflation	INFL	Consumer price index for all urban consumers.
Term spread	TMS	Difference between the long-term yield and the Treasury bill rate.

Note: This table presents 14 candidate predictors used for forecasting market excess returns and provides a brief description of their respective measurement methodologies. These predictors are sourced from [Goyal and Welch \(2008\)](#). In our analysis, we compare the predictive performance of our proposed *PWI* against these variables using in-sample R-squared, out-of-sample R-squared, and the effectiveness of investment strategies.

Table 3: Summary Statistics of Candidate Predictor Variables

Variable	Mean	SD	Min	Median	Max
PWI	1.0349	0.2802	0.6408	0.9784	2.6758
BM	0.2854	0.0602	0.1756	0.2935	0.4242
TBL	0.0086	0.0131	0.0001	0.0014	0.0530
LTY	0.0262	0.0086	0.0062	0.0265	0.0432
NTIS	-0.0065	0.0141	-0.0326	-0.0097	0.0201
INFL	0.0021	0.0035	-0.0067	0.0019	0.0137
LTR	0.0037	0.0290	-0.0629	-0.0003	0.0862
SVAR	0.0026	0.0061	0.0002	0.0013	0.0732
LOG_DP	-3.9836	0.1314	-4.3684	-3.9496	-3.7700
LOG_DY	-3.9747	0.1317	-4.3597	-3.9387	-3.7694
LOG_EP	-3.0517	0.2135	-3.6865	-3.0860	-2.5656
LOG_DE	-0.9319	0.1915	-1.2442	-0.9466	-0.4794
TMS	0.0176	0.0132	-0.0157	0.0186	0.0419
DFY	0.0096	0.0023	0.0055	0.0093	0.0170
DFS	0.0010	0.0182	-0.0976	0.0009	0.0426

Note: This table presents the summary statistics for the candidate predictors and our proposed *PWI* predictor over the period 2010–2023. *PWI* is initially estimated on a weekly basis and subsequently converted to a monthly frequency by taking the average. The maximum observed value of *PWI* is 2.6758, while the minimum is 0.6408.

Table 4: Fitting Performance of Different Volatility Models

	Coefficient	t-Statistics	Log-likelihood	AIC
ARCH	$\mu = 0.0008$	5.10	10898.69	-21791.38
	$\omega = 7.32E - 05$	36.36		
	$\alpha = 0.3856$	10.99		
GARCH	$\mu = 0.0008$	6.10	11347.75	-22687.49
	$\omega = 3.66E - 06$	7.02		
	$\alpha = 0.1708$	10.25		
	$\beta = 0.7989$	47.76		
EGARCH	$\mu = 0.0004$	2.96	11414.39	-22818.78
	$\nu = -0.4378$	-14.45		
	$\beta = 0.9530$	296.66		
	$\zeta = 0.2215$	12.05		
	$\kappa = -0.1591$	-18.04		

Note: This table reports the fitting performance of three different volatility models: ARCH, GARCH, and EGARCH. The table presents the estimated coefficients for each model along with their corresponding t-statistics, as well as the log-likelihood and Akaike Information Criterion (AIC) values, which are used to evaluate model fit.

Table 5: In-sample Estimation Results: Univariate Predictive Regressions

	β	t -stat	$R^2(\%)$		β	t -stat	$R^2(\%)$
PWI	0.0058*	1.70	1.9	LOG_DP	0.0079**	2.47	3.5
BM	0.0059*	1.93	2.0	LOG_DY	0.0054*	1.88	1.7
TBL	-0.0025	-0.87	0.4	LOG_EP	0.0018	0.53	0.2
LTY	-0.0034	-1.00	0.6	LOG_DE	0.0034	0.99	0.6
NTIS	0.0012	0.46	0.1	TMS	0.0003	0.08	0.0
INFL	-0.0031	-1.21	0.5	DFY	0.0046	1.57	1.2
LTR	0.0019	0.42	0.2	DFS	-0.0013	-0.29	0.1
SVAR	0.0075***	2.95	3.2				

Note: This table reports the in-sample estimation results for Equation (3.23). All predictive variables are standardized. The table shows parameter estimates, corresponding Newey-West t -statistics, and in-sample R^2 values (in percentage). Asterisks *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 6: In-sample Estimation Results: Bivariate Predictive Regressions

Variable	β_{PWI}	β_{GW}	t_{PWI}	t_{GW}	$R^2(\%)$
BM	0.0090***	0.0091***	2.85	2.69	6.1
TBL	0.0063*	-0.0033	1.86	-1.06	2.6
LTY	0.0054	-0.0008	1.35	-0.22	2.0
NTIS	0.0061*	0.0021	1.79	0.81	2.2
INFL	0.0057*	-0.0028	1.69	-1.09	2.4
LTR	0.0057	0.0012	1.64	0.26	2.0
SVAR	0.0024	0.0062*	0.65	1.71	3.4
LOG_DP	0.0066**	0.0084***	2.26	2.64	6.0
LOG_DY	0.0065**	0.0061**	2.12	2.08	4.0
LOG_EP	0.0079**	0.0051	2.19	1.42	3.2
LOG_DE	0.0053	0.0012	1.34	0.31	2.0
TMS	0.0073*	0.0034	1.83	0.88	2.5
DFY	0.0049	0.0031	1.38	1.07	2.4
DFS	0.0058*	-0.0004	1.76	-0.09	2.0

Note: This table reports the in-sample estimation results for Equation (3.24). All predictive variables are standardized. The table displays parameter estimates, their corresponding Newey-West t-statistics, and in-sample R^2 values (in percentage). Asterisks *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 7: Out-of-Sample Forecasting Results of Univariate Models

	In-Sample	Out-of-Sample	All Sample		In-Sample	Out-of-Sample	All Sample
PWI	0.33	2.18	1.27	LOG_DP	9.46	-43.27	-17.42
BM	2.42	0.96	1.67	LOG_DY	3.17	-11.59	-4.36
TBL	0.15	0.58	0.37	LOG_EP	2.07	-4.03	-1.04
LTY	0.92	-0.11	0.40	LOG_DE	0.15	-1.54	-0.71
NTIS	0.33	-0.28	0.01	TMS	0.06	-0.31	-0.13
INFL	0.60	0.38	0.58	DFY	0.33	1.25	0.80
LTR	2.51	-3.44	-0.52	DFS	0.07	-0.45	-0.20
SVAR	0.30	-11.47	-5.70	AVG	1.61	-5.24	-1.88

Note: This table presents the out-of-sample forecasting performance of univariate models using the split-sample approach. The models are first estimated in the in-sample period based on Equation (3.23) and then applied to predict returns in the out-of-sample period. The forecast accuracy is assessed using the out-of-sample R^2 , as defined in Equation (3.26). Higher values of R^2 indicate better predictive performance. The column "All Sample" represents the full-period estimation results.

Table 8: Out-of-Sample $R^2(\%)$ Using Expanding Window Approach

Predictor	1-Month	2-Month	3-Month	6-Month
PWI	-1.16	2.51	-6.35	-12.62
BM	-2.26	0.20	-15.20	-22.23
TBL	-5.45	-0.66	-12.91	-37.24
LTY	-1.91	3.52	-6.75	-23.41
NTIS	-0.95	4.21	-3.89	-9.48
INFL	-2.10	6.83	-9.17	-5.20
LTR	-1.40	7.48	1.69	-6.70
SVAR	-15.20	0.27	-11.05	-28.73
LOG_DP	-1.22	-8.20	-33.86	-16.74
LOG_DY	-1.89	-3.48	-19.94	-1.38
LOG_EP	-2.69	0.36	-13.28	-33.49
LOG_DE	-2.20	4.55	-3.70	-30.01
TMS	-2.97	1.20	-6.02	-18.70
DFY	-0.68	5.33	-0.40	7.26
DFS	-3.47	7.30	-1.75	-5.24
AVG	-3.17	2.07	-9.73	-16.52

Note: This table presents the out-of-sample predictive performance of univariate models using the expanding window approach. The estimation window gradually increases over time, incorporating additional data as new observations become available. The forecast quality is assessed using the out-of-sample R^2 , as defined in Equation (3.26). The table reports results across different forecast horizons: 1-month, 2-month, 3-month, and 6-month. A higher R^2 value indicates better predictive accuracy.

Table 9: Out-of-Sample Performance of Univariate Models: CEQ Gains (%)

Model	RA = 1	RA = 2	RA = 3	AVG
Buy & Hold	7.72	6.62	5.52	6.62
PWI	9.97	7.59	4.76	7.44
BM	3.73	2.98	1.88	2.86
TBL	6.55	4.53	2.14	4.41
LTY	8.16	6.64	5.09	6.63
NTIS	9.89	7.12	3.86	6.96
INFL	9.15	5.98	4.41	6.51
LTR	9.56	7.17	5.47	7.40
SVAR	6.65	4.41	2.06	4.37
LOG_DP	10.51	8.19	6.59	8.43
LOG_DY	9.79	8.63	6.51	8.31
LOG_EP	3.43	1.38	0.28	1.70
LOG_DE	7.37	5.28	3.55	5.40
TMS	9.63	6.22	3.54	6.46
DFY	9.47	4.82	3.43	5.91
DFS	8.45	5.37	3.50	5.77
AVG	8.02	5.62	3.74	5.79

Note: This table presents the out-of-sample performance of investment strategies based on univariate predictive models. The evaluation metric is the certainty equivalent return (CEQ), which represents the annualized risk-adjusted return for an investor with constant relative risk aversion (RA). The CEQ is reported in percentage terms and is calculated for three levels of risk aversion: RA = 1, RA = 2, and RA = 3. The last column (AVG) represents the average CEQ across these risk aversion levels. A higher CEQ indicates superior investment performance.

Table 10: Out-of-Sample Performance of Bivariate Models: CEQ Gains (%)

Predictor	RA = 1	RA = 2	RA = 3	AVG
BM	5.40	2.89	1.71	3.33
TBL	5.78	3.56	1.19	3.51
LTY	9.49	6.81	3.90	6.73
NTIS	9.01	7.14	4.26	6.80
INFL	8.89	6.13	3.86	6.29
LTR	9.70	7.81	5.45	7.65
SVAR	7.17	4.59	2.71	4.82
LOG_DP	8.41	7.18	5.51	7.03
LOG_DY	9.69	6.86	5.76	7.44
LOG_EP	4.51	1.74	0.81	2.35
LOG_DE	9.21	6.54	4.31	6.69
TMS	10.04	6.55	4.07	6.89
DFY	9.21	5.21	2.63	5.68
DFS	7.61	4.76	2.57	4.98

Note: This table presents the out-of-sample performance of investment strategies based on bivariate predictive regressions. Each model includes one of the predictors from Goyal and Welch (2008) alongside our proposed *PWI* predictor. The performance metric is the certainty equivalent return (CEQ), which represents the annualized risk-adjusted return for investors with different levels of relative risk aversion ($RA = 1$, $RA = 2$, and $RA = 3$). The last column (AVG) reports the average CEQ across the three risk aversion levels. An increase in CEQ after incorporating *PWI* suggests that it enhances predictive power and improves investment performance.

Table 11: Out-of-Sample Performance of Forecast Combinations: CEQ Gains (%)

	RA = 1		RA = 2		RA = 3		AVG	
	GW	PWI	GW	PWI	GW	PWI	GW	PWI
Mean	8.09	8.60	5.34	5.76	3.35	3.59	5.59	5.98
Trimmed Mean	8.18	8.77	5.50	6.02	3.35	3.84	5.68	6.21
DMSPE (0.9)	8.19	8.73	5.20	5.58	2.99	3.28	5.46	5.86
DMSPE (1)	8.21	8.74	5.26	5.69	3.15	3.42	5.54	5.95
Yang	8.39	8.90	5.56	5.98	3.50	3.74	5.82	6.21

Note: This table presents the out-of-sample performance of investment strategies based on forecast combinations. Using the combined forecast and Equation (3.27), we determine the optimal portfolio allocation between the risky asset and the risk-free asset, employing the same methodology as in the univariate and bivariate predictive models. The table reports results for five different forecast combination approaches: Mean, Trimmed Mean, Discounted Mean Squared Prediction Error (DMSPE) with discount factors of 0.9 and 1, and Yang's method. For each approach, we compare the certainty equivalent return (CEQ) before and after incorporating the *PWI* predictor. Results are presented for three levels of relative risk aversion ($RA = 1$, $RA = 2$, and $RA = 3$), with the final two columns reporting the average CEQ across all risk aversion levels. An increase in CEQ when *PWI* is included suggests that it enhances the predictive power of the forecast combination.

Table 12: Forecasting Cash Flows and Discount Rates with *PWI*

Variable	β	<i>t</i> -stat	$R^2(\%)$
DP	-0.0308***	-2.881	90.00
LTY	-0.8019***	-3.104	9.50
DG	-0.0389	-0.260	0.10
EG	-0.0872*	-1.793	15.00

Note: This table presents the estimation results of Equation (3.33), examining the predictive relationship between *PWI* and proxies for cash flow and discount rate dynamics. The dependent variables include the dividend-price ratio (DP), long-term yield (LTY), dividend growth (DG), and earnings growth (EG). The coefficients (β) measure the impact of *PWI* on these variables, with *t*-statistics indicating statistical significance. Adjusted R^2 values (in percentage) reflect the explanatory power of each regression. The *t*-statistic is calculated based on the Newey-West method. Asterisks *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.